

The Theory of Persistence: geometric and spectral readings

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Abstract

The Theory of Persistence (PT), exposed in its companion article [1], derives from a single arithmetic axiom $s = 1/2$ a cascade structure on the active primes $\{3, 5, 7\}$ and an arithmetic fixed point $\mu^* = 15$. This second article explores three complementary readings of that structure: (i) an *algebro-geometric* reading that exhibits the canonical cascade as the unique algebraic curve in the Kontsevich–Norbury class satisfying three conditions of holonomy, activation, and self-consistency, and computes its genus $g = \mu^* - 1 = 14$; (ii) a *differential* reading that establishes that the Fisher–Bianchi metric induced by the cascade satisfies the Ricci soliton equation $\text{Ric} + \text{Hess}(f) = \lambda(\mu) g_{\text{PT}}$ with $\lambda(\mu) = -1/\mu^4 + O(1/\mu^5)$, coefficient -1 exact; (iii) a *spectral* reading that establishes the triple identity $1/8 = s^2/2 = \lambda_H = \Delta_N^{\text{Maslov}}$ relating the Higgs coupling and the Maslov phase of the Riemann zeros to the unique formal input of PT, together with an exact trace identity verified to 10^{-26} . The more ambitious spectral-arithmetic program whose first stones these identities are (RH-PT conjecture) is sketched as perspective.

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1 Introduction and reminders

1.1 Why geometric and spectral readings?

The companion article [1] exposes the purely arithmetic foundations of the Theory of Persistence (PT): a single axiom $s = 1/2$, seven structural theorems **T0–T6**, three proved bridge axioms **BA3–BA5**, a fixed point $\mu^* = 15$, and a canonical cascade on the active primes $\{3, 5, 7\}$. This structure produces, without adjustable parameters, physical observables such as the fine-structure constant α_{EM} at 0,004 ppb.

Once this arithmetic basis is in place, three natural questions arise.

1. *Is the cascade an algebro-geometric object?* Does the sequence $\gamma_3 \rightarrow \gamma_3\gamma_5 \rightarrow \gamma_3\gamma_5\gamma_7$ admit a realisation as an algebraic curve whose topological invariant encodes μ^* ?
2. *Is the cascade a differential-geometric object?* Does the information structure induce a canonical Riemannian metric on the parameter space, and does that metric satisfy a Ricci equation?
3. *Is the cascade a spectral object?* Does the axiom $s = 1/2$, which fixes the dominant eigenvalue of the primordial transfer matrix (**T2**), produce identifiable spectral signatures in physical phenomenology or in the structure of the Riemann zeros?

This article answers all three questions positively, presenting one principal result per reading.

1.2 Minimal reminders

The reader unfamiliar with [1] will find below the elements strictly necessary for what follows. For justifications, we refer to the companion article.

Axiom and fixed point. The unique formal input of PT is $s = 1/2$. The arithmetic fixed point of the primordial sieve, determined by theorem **T5**, is

$$\mu^* = 3 + 5 + 7 = 15.$$

The set of primes splits canonically into four classes:

$$\underbrace{\{3, 5, 7\}}_{\text{active}}, \quad \underbrace{\{11, 13\}}_{\text{echoes}}, \quad \underbrace{\{17, 19, 23\}}_{\text{super-echoes}}, \quad \underbrace{\{p \geq 29\}}_{\text{inactive}},$$

plus the parity channel $\{2\}$.

Canonical cascade. The canonical PT cascade ([1, §5]) is the ordered sequence

$$\mathcal{C}_0 \xrightarrow{\gamma_3} \mathcal{C}_3 \xrightarrow{\gamma_5} \mathcal{C}_{3,5} \xrightarrow{\gamma_7} \mathcal{C}_{3,5,7},$$

where $\gamma_p(\mu) = 1 - 2/(p(1 - q^p))$, $q = 1 - 2/\mu$, are the *anomalous dimensions*. At $\mu^* = 15$, we have $\gamma_3 = 0,808$, $\gamma_5 = 0,696$, $\gamma_7 = 0,595$.

Holonomy and sectors. The holonomy formula (BA3) gives, for each active prime,

$$\sin^2 \theta_p(q) = \delta_p(q) (2 - \delta_p(q)), \quad \delta_p(q) = \frac{1 - q^p}{p}.$$

The two canonical parametrisations of the geometric distribution at the fixed point are $q_+ = 13/15$ (vertex branch, rational, max-entropy) and $q_- = e^{-1/15}$ (edge branch, transcendent, Gibbs), related by an involution D .

Pontryagin product. The sieve coupling functional, at the fixed point μ^* , takes the product form (BA5),

$$\alpha_{\text{sieve}} = \prod_{p \in \{3,5,7\}} \sin^2 \theta_p(q_+).$$

1.3 Plan, epistemic status

What follows is organised into five complementary sections.

- Section 2 establishes the *persistence curve theorem*: the canonical cascade admits a unique algebraic curve in the Kontsevich–Norbury class satisfying three conditions of holonomy, threshold, and self-consistency. Its genus is $g = \mu^* - 1 = 14$.
- Section 3 establishes the *Ricci quasi-soliton theorem*: the Fisher–Bianchi metric g_{PT} induced by the cascade satisfies $\text{Ric}(g_{\text{PT}}) + \text{Hess}(f) = \lambda(\mu) g_{\text{PT}}$ with $\lambda(\mu) = -1/\mu^4 + O(1/\mu^5)$, coefficient -1 exact.
- Section 4 proves the *spectral identity K2*: $1/8 = s^2/2 = \lambda_H = \Delta_N^{\text{Maslov}}$, identifying the Maslov phase of the Riemann zeros with the Higgs coupling and the PT axiom, and the *trace identity A6* verified numerically to 10^{-26} .
- Section 5 sketches the more ambitious spectral-arithmetic program (*RH-PT conjecture*) whose first stones are the identities K2 and A6.
- Section 6 synthesises the three readings and outlines perspectives.

Epistemic status. The results of this article fall into three distinct categories that should not be conflated.

Theorems proved unconditionally : persistence curve and genus computation (2, from [2]), asymptotic Ricci quasi-soliton (3, from [3]).

Proved identities : K2 and A6 are exact identities (the first structurally, verified to all available precisions; the second algebraically in $\Re(s) > 1$, verified numerically to 10^{-26}). Their extension to the critical line remains open.

Conjectural program : the RH-PT conjecture, which would extend A6 and K2 to $\Re(s) = 1/2$, is an active research program, not a theorem.

This differentiation is maintained throughout the article.

2 Algebro-geometric reading: the persistence curve

The canonical PT cascade (1.2) is a purely combinatorial sequence of arithmetic refinements. This section shows that it admits a unique algebro-geometric realisation: there exists an algebraic curve in the Kontsevich–Norbury class which encodes exactly the active-prime cascade, and its fundamental topological invariant — the genus — is given by the remarkable formula $g = \mu^* - 1 = 14$. All results in this section are rigorously proved in [2]; we give the statements and proof ideas.

2.1 Preliminaries: Kontsevich–Norbury spectral curves

A *spectral curve* in the sense of the Eynard–Orantin topological recursion [4] is a quadruple (Σ, x, y, B) where Σ is a Riemann surface, x and y are two meromorphic functions on Σ , and

B is a bilinear form (the *Bergman kernel*). The Kontsevich–Norbury class [5, 6], denoted $\mathcal{KN}$, is the subclass of spectral curves whose univariate form $\omega_{0,1} = y dx$ determines, via topological recursion, the full set of Weil–Petersson invariants of the moduli space $\overline{\mathcal{M}}_{g,n}$ of pointed Riemann surfaces. The archetypical case is Mirzakhani’s Airy curve [7], whose univariate form exhibits the constant $2\pi^2/3$ characteristic of Weil–Petersson volumes.

2.2 Three self-consistency conditions

To a cascade $S = \{p_{(1)}, \dots, p_{(k)}\} \subseteq \mathbb{N}^*$ with $\mu^*(S) := \sum_{p \in S} p > 2$ we naturally associate a family of candidate curves in $\mathcal{KN}$, parametrised by a complex variable z via

$$\mu_S(z) = \frac{\mu^*(S)}{1 - z^2}, \quad q_S(z) = 1 - \frac{2}{\mu_S(z)}, \quad (1)$$

and the function y defined by the cascade product

$$y_S(z) = \frac{z}{2\pi} \prod_{n \in S} \sin \theta_n(\mu_S(z)), \quad \omega_{0,1}^{C_S} = y_S(z) z dz. \quad (2)$$

The parametrisation (1) is rational even in z , has simple poles at $z = \pm 1$, and satisfies $\mu_S(0) = \mu^*(S)$. The point $z = 0$ is therefore the *self-consistency point*: the value of the parameter μ there is precisely the arithmetic fixed point.

For such a curve to authentically encode the PT cascade, three structural conditions must be simultaneously imposed.

Definition 2.1 (ACA $_{\mathcal{N}}$ cascade conditions). Let $\mathcal{N} \subseteq \mathbb{N}^*$ be a sieve (typically $\mathcal{N} = \{\text{primes}\}$, but other choices are admissible), and let (S, μ) with $S \subseteq \mathcal{N}$ finite and $\mu : \mathbb{C} \rightarrow \widehat{\mathbb{C}}$ rational even.

- (A) *Polynomial holonomy*. For each $n \in S$, the cyclic phase satisfies $\sin^2 \theta_n = \delta_n(q)(2 - \delta_n(q))$ with $\delta_n(q) = (1 - q^n)/n$ (PT bridge axiom **BA3**).
- (B) *Activation threshold*. The subset S coincides with the elements of $\mathcal{N}$ for which $\gamma_n(\mu^*) > 1/2$ (axiom **BA4**).
- (C) *Self-consistency*. The parameter satisfies $\mu(0) = \mu^*(S) := \sum_{n \in S} n$ (theorem **T5**).

On the prime sieve $\mathcal{N} = \mathcal{P}$, conditions (A), (B), (C) are precisely the algebraic translation of Article 1’s theorems and bridge axioms. The principal theorem of this section asserts that these three conditions select a *unique* subset S and that the associated curve is algebraic.

2.3 Uniqueness theorem and classification

Theorem 2.2 (Universal classification, simplified version). *Let $\mathcal{P}$ be the set of prime numbers. Among the subsets $S \subseteq \mathcal{P}$ satisfying conditions ACA $_{\mathcal{P}}$ (2.1), there is exactly one S with $|S| \geq 2$ and $\mu^*(S) \leq 200$: it is*

$$S = \{3, 5, 7\}, \quad \mu^*(S) = 15.$$

Idea of proof. Condition (B), $\gamma_n(\mu^*) > 1/2$, is highly constraining: combined with the decreasing monotonicity of $\gamma_n(\mu)$ with n proved in the companion article [8], it forces S to be contained in an initial segment of the primes. Condition (C), $\mu^*(S) = \sum_{n \in S} n$, then balances the effect: an S too small fails to saturate the sum, an S too large violates the threshold. Exhaustive numerical search over all 2406 compatible subsets up to $\mu^* \leq 200$ identifies $S = \{3, 5, 7\}$ as the unique solution with $|S| \geq 2$ [2, §5]. The analytic proof completes the argument for $\mu^* > 200$ via prime number theorem bounds.

2.4 Explicit algebraic form

Theorem 2.3 (Algebraicity of the persistence curve). *The curve $\mathcal{C}_S$ defined by (1)–(2) with $S = \{3, 5, 7\}$ is algebraic: the function y^2 is a rational fraction of $x = z^2/2$. Explicitly, setting $q(x) = (\mu^* - 2 + 4x)/\mu^*$,*

$$Y^2 = \frac{2x}{4\pi^2} \prod_{n \in S} \frac{(1 - q(x)^n)(2n - 1 + q(x)^n)}{n^2} = \frac{N_S(x)}{2\pi^2 D_S}, \quad (3)$$

with $D_S \in \mathbb{N}$ and $N_S \in \mathbb{Z}[x]$ of degree $2\mu^* + 1$. The polynomial N_S has x as factor (multiplicity 1, from the prefactor $z^2 = 2x$) and $(2x - 1)$ as factor (multiplicity $|S|$, simultaneous vanishing of the $|S|$ factors $(1 - q^n)$ at $q = 1$).

Idea of proof. Since $\mu_S(z)$ is rational even, $q_S(z) = 1 - 2/\mu_S(z)$ is too. Each $\sin^2 \theta_n(\mu_S(z)) = \delta_n(2 - \delta_n)$ with $\delta_n = (1 - q^n)/n$ is rational in q^n , hence in z^2 , hence in $x = z^2/2$. The product (3) is therefore rational in x . For $S = \{3, 5, 7\}$, the explicit degree of N_S is $2 \cdot 15 + 1 = 31$; exact symbolic computation (Python/SymPy, scripts in [2]) confirms the factor structure announced.

2.5 Genus computation and topological consequence

Theorem 2.4 (Universal genus formula). *Let $S \subseteq \mathcal{P}$ be a solution of conditions $ACA_{\mathcal{P}}$ satisfying the hypotheses of 2.2. Then the genus of the algebraic curve underlying $\mathcal{C}_S$ is*

$$g(\mathcal{C}_S) = \mu^*(S) - \lfloor |S|/2 \rfloor.$$

Consequence for PT. Applying the formula to the unique solution $S = \{3, 5, 7\}$ of 2.2,

$$g(\mathcal{C}_{\text{per}}) = \mu^* - \lfloor 3/2 \rfloor = 15 - 1 = 14. \quad (4)$$

Idea of proof. From the explicit form (3) we extract the squarefree factor of N_S : $N_S^{\text{sqf}}(x) = N_S(x)/(2x - 1)^{|S|-1}$. Its degree is $2\mu^* + 1 - (|S| - 1) = 2\mu^* - |S| + 2$. The hyperelliptic curve $Y^2 = N_S^{\text{sqf}}(x)$ then has genus $\lfloor (\deg N_S^{\text{sqf}} - 1)/2 \rfloor = \lfloor (2\mu^* - |S| + 1)/2 \rfloor = \mu^* - \lfloor |S|/2 \rfloor$. For $S = \{3, 5, 7\}$, direct computation: $g = 15 - 1 = 14$. The exhaustive computation of discriminants of squarefree polynomials is in [2, §7].

2.6 Interpretation: μ^* as a topological invariant

Identity (4) transforms the arithmetic fixed point $\mu^* = 15$ from a mere integer into a *topological invariant*: it is the genus, plus one, of a Riemann surface canonically associated to the cascade. This translation carries three important consequences.

1. The triplet $\{3, 5, 7\}$ is not merely compatible with the equation $\mu = \sum_{p \in S} p$: it is the unique support of a non-trivial algebraic curve in the Kontsevich–Norbury class. The rigidity of the active-prime choice receives a *topological* justification, complementary to the arithmetic justification by axioms **BA3–BA4**.
2. The PT cascade fits into the general framework of Eynard–Orantin topological recursion: the associated invariants $\omega_{g,n}^{\text{C}_{\text{per}}}$ inherit classical identities (Virasoro relations, intersection formulas on $\mathcal{M}_{g,n}$). A structural substitution Mirzakhani $\rightarrow$ persistence, $2\pi^2/3 \rightarrow (1/2) \sum_{p \in S} \gamma_p$, transports Weil–Petersson volumes towards PT invariants [2, §8].
3. Genus 14 is, to the author’s knowledge, the first strictly non-trivial topological invariant attached to the arithmetic structure of PT, opening the way to a cohomological study of its applications.

The next section develops a second, complementary reading equipping the parameter space μ with a canonical Riemannian metric satisfying a Ricci soliton equation.

3 Differential reading: the Ricci quasi-soliton

The PT arithmetic cascade naturally induces, on the parameter space μ , a canonical Riemannian metric. This section establishes that this metric satisfies, asymptotically, the Ricci soliton equation of Hamilton and Perelman, with an explicit and exact coefficient. PT thereby fits into the broad framework of modern differential geometry, and its arithmetic fixed point μ^* receives a geometric reading complementary to the algebraic one in 2. All results are rigorously proved in [3].

3.1 The Fisher–Bianchi metric induced by the cascade

Let $\mu > 0$ be the arithmetic parameter and $q = q(\mu) = 1 - 2/\mu$ the vertex-branch parameter. For each active prime $p \in S^* := \{3, 5, 7\}$, define the *local Hubble rate*

$$H_p(\mu) = -\frac{1}{2\mu^2} \frac{p q^{p-1}}{1 - q^p}, \quad (5)$$

measuring the rate of change of the holonomy phase at prime p with μ . The *Fisher–Bianchi metric of PT*, denoted g_{PT} , is the Riemannian metric on the model temporal manifold $\mathbb{R} \times \mathbb{R}^{|S^*|}$ defined, in the gauge $N(\mu) = \mu$, by

$$g_{\text{PT}} = -d\mu^2 + \sum_{p \in S^*} a_p(\mu)^2 dx_p^2, \quad a_p(\mu) := \exp\left(\int H_p(\mu) d\mu\right). \quad (6)$$

This metric is the PT analogue of a Lorentzian Bianchi I metric [9], but its scale factors $a_p(\mu)$ are *not isotropic*: they differ from one active prime to another, which distinguishes g_{PT} from a Robertson–Walker metric. Precisely this anisotropy will permit the determination of a unique Ricci residue.

The introduction of metric (6) is canonical in the following sense: its scale factors derive directly from the holonomy formula **BA3**, and the gauge $N(\mu) = \mu$ is the only one that renders the entropy potential $S = -\ln \alpha_{\text{sieve}}$ a *regular* function of the time coordinate ([3, §2]).

3.2 The Ricci soliton equation

Let f be a smooth function on the temporal manifold. The *gradient Ricci soliton* equation of Hamilton and Perelman [10, 11] reads

$$\text{Ric}(g) + \text{Hess}(f) = \lambda g, \quad \lambda \in \mathbb{R}. \quad (7)$$

Stationary solutions ($\lambda = 0$, said *steady*), expanding ($\lambda < 0$), and shrinking ($\lambda > 0$) classify the asymptotic regimes of evolution under the Ricci flow $\partial_t g = -2 \text{Ric}(g)$. Perelman’s geometry systematically studies these solutions and their singularities, in particular in the proof of the Poincaré conjecture [11].

In the PT setting, we choose $f := S = -\ln \alpha_{\text{sieve}}$: the potential function is the negative log of the Pontryagin coupling. Equation (7), projected onto the purely spatial components (pp) , $p \in S^*$, gives three equations in two unknowns $(\dot{f}, \lambda)$. The main theorem exploits this overdetermination.

3.3 Asymptotic expanding quasi-soliton theorem

Theorem 3.1 (Asymptotic Ricci quasi-soliton). *Let g_{PT} be the PT Fisher–Bianchi metric (6) in the gauge $N(\mu) = \mu$, and let $f = S = -\ln \alpha_{\text{sieve}}(\mu)$. Imposing compatibility of the three (pp) equations of the soliton equation (7) to determine $\lambda(\mu)$, we have*

$$\lambda(\mu) = -\frac{1}{\mu^4} + O\left(\frac{1}{\mu^5}\right) \quad \text{as } \mu \rightarrow \infty. \quad (8)$$

The coefficient -1 of the leading term in μ^{-4} is exact and independent of the choice of pair (p_1, p_2) of active primes used to solve the system.

Immediate consequence. $\lambda < 0$ asymptotically: the PT Fisher–Bianchi metric is an *expanding quasi-soliton* in the sense of Hamilton–Perelman. The qualifier “quasi-” refers to the fact that $\lambda(\mu) \rightarrow 0$ as $\mu \rightarrow \infty$: a strict steady soliton would have $\lambda \equiv 0$, and a strict expanding soliton would have λ constant non-zero; PT belongs to an intermediate regime whose deviation from $\lambda = 0$ decreases as μ^{-4} .

3.4 Proof sketch: anisotropy–uniqueness equivalence

The determination of $\lambda(\mu)$ rests on a structural equivalence between effective anisotropy of Hubble rates and uniqueness of the λ solution.

Proposition 3.2 (Anisotropy $\Leftrightarrow$ uniqueness of λ). *For the metric g_{PT} and any $\mu > 0$, the following two properties are equivalent:*

- (i) *There exist $p_1 \neq p_2$ in S^* such that $H_{p_1}(\mu) \neq H_{p_2}(\mu)$ (effective anisotropy).*
- (ii) *The system of (pp) equations for $p \in S^*$ admits a unique solution in the unknown λ .*

Proof. When the H_p are all equal, the system of three (pp) equations reduces to a single equation in two unknowns $(\dot{f}, \lambda)$: λ is then a free degree of freedom. As soon as a single pair (p_1, p_2) with $H_{p_1} \neq H_{p_2}$ appears, the difference of the two equations $(p_1 p_1) - (p_2 p_2)$ eliminates $\dot{f}$ and provides λ explicitly [3, §4]. $\square$

Lemma 3.3 (Automatic asymptotic anisotropy). *For $p_1 \neq p_2$ in S^* ,*

$$H_{p_1}(\mu) - H_{p_2}(\mu) = \frac{p_2 - p_1}{\mu^3} + O(\mu^{-4}) \neq 0 \quad \text{for all } \mu \text{ sufficiently large.}$$

3.1 then follows directly: for $\mu \rightarrow \infty$, effective anisotropy is automatic (lemma), hence λ is unique (proposition), and computing the leading term of the difference of the (pp) equations gives $\lambda(\mu) = -1/\mu^4 + O(\mu^{-5})$. The independence of the coefficient -1 from the choice of (p_1, p_2) results from the system’s structure: any choice of pair produces the same linear equation after elimination of $\dot{f}$, up to terms of order μ^{-5} .

3.5 Interpretation: residue, dynamics, and link to Perelman

The residue as a signature of non-trivial dynamics. A strict steady soliton would satisfy $\text{Ric} + \text{Hess}(f) = 0$, that is, the metric would be invariant under the Ricci flow (up to diffeomorphism and reparametrisation). The non-zero residue $\lambda(\mu) \sim -1/\mu^4$ of PT shows that the arithmetic cascade is *not* in such an exact equilibrium regime: it evolves slowly towards equilibrium, at a rate decreasing with sieve depth. This is a geometric signature of *non-trivial dynamics*; by cosmological analogy, the residue can be read as an internal “relaxation time” of the cascade.

Inscription in the Hamilton–Perelman program. Ricci solitons are the central objects of Hamilton’s Ricci-flow program, formalised and completed by Perelman [11]. They serve in the classification of Ricci flow singularities and in the proof of the Poincaré and Thurston geometrization conjectures. Theorem 3.1 thus places PT within this landscape: the arithmetic cascade of active primes appears no longer as a mere count but as a *differential-geometric structure* satisfying a canonical evolution equation.

Articulation with the algebraic reading. The two readings of the PT object — algebraic (2) and differential (this section) — do not merely coexist: they are related by a *scale bridge* proved in [3, §6]:

$$\lambda(\mu) \sim B_0(\mu)^{-8/3}, \quad \mu \rightarrow \infty,$$

where $B_0(\mu)$ is the leading coefficient of the topological-recursion invariant $\omega_{1,1}^{\text{C}_{\text{per}}}$ of the persistence curve. The exponent $-8/3$ is verified numerically to 0,4%; its *exact* identification remains an open question that constitutes one of the goals of the spectral program presented in 5.

With these two readings in place, the next section develops a third: the spectral reading, which relates the arithmetic axiom $s = 1/2$ directly to the spectral signatures of the Higgs coupling and the Riemann zeros.

4 Spectral reading: identities K2 and A6

The two previous readings (algebraic in 2, differential in 3) show that the PT arithmetic cascade admits canonical geometric realisations. The present section establishes a third reading: the cascade leaves *spectral signatures* in two classical structures of pure mathematics and theoretical physics — the distribution of zeros of Riemann’s zeta function on one hand, and the Higgs coupling of the Standard Model on the other. The principal result (identity K2) is a triple identification of the Riemann–von Mangoldt residue as an expression of the arithmetic axiom $s = 1/2$; the auxiliary result (trace identity A6) is an exact formula verified numerically to 10^{-25} . These results stem from the consolidation [12].

4.1 Density of Riemann zeros and the $1/8$ residue

Let $N(T)$ denote the number of non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \gamma < T$. The classical Riemann–von Mangoldt theorem [13] provides the asymptotic expansion

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8} + S(T) + O(1/T), \quad (9)$$

where $S(T) = \frac{1}{\pi} \arg \zeta(1/2 + iT)$ is the argument function and oscillates with zero mean. The constant $7/8$ is a fine spectral signature: in the Berry–Keating semi-classical quantisation [14], it encodes a *Maslov phase* associated with the turning points of the regularised Hamiltonian model $H_{BK} = xp$.

PT predicts [12] a semi-classical variant of this formula, in which $7/8 = 1 - 1/8$ is replaced exactly by 1:

$$N_{PT}(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + 1 + S(T) + O(1/T). \quad (10)$$

The gap between (9) and (10) is exactly $\Delta_N = 7/8 - 1 = -1/8$, or in absolute value, $|\Delta_N| = 1/8$. The question posed by identity K2 is: does this $1/8$ residue have a pure PT reading?

4.2 Identity K2: triple identification of the $1/8$ residue

Theorem 4.1 (K2 identity, spectral-physical identification). *The absolute residue $|\Delta_N| = 1/8$ admits, at the fixed point $\mu^* = 15$ of the PT cascade, three structurally equivalent exact identifications:*

$$\boxed{\frac{1}{8} = \frac{s^2}{2} = \lambda_H = \Delta_N^{\text{Maslov}},} \quad (11)$$

where λ_H is the Higgs self-coupling of the Standard Model (result **R lambda_H** of the companion article [15]), and Δ_N^{Maslov} is the Maslov phase of the double-barrier corners in the regularised Berry–Keating model.

Three identifications. Identity (11) decomposes into three exact sub-identities, each derivable independently in the PT framework.

P1. *Square of the symmetry parameter.* The axiom $s = 1/2$ gives directly

$$\frac{s^2}{2} = \frac{1/4}{2} = \frac{1}{8}.$$

P2. Higgs self-coupling. The Higgs scalar coupling derivation developed in the companion article [15] provides, at cascade depth $d = 3$,

$$\lambda_H = \frac{s^2}{2} = \frac{1}{8}.$$

This identity is obtained independently of any spectral consideration, via the self-coupling structure of the scalar potential at the fixed point.

P3. Maslov phase. The regularised Berry–Keating model for the Riemann zeros gives, for the Maslov phase of the double-barrier corners [14],

$$\Delta_N^{\text{Maslov}} = \frac{1}{8} = -\frac{\pi/8}{\pi/1},$$

in semi-classical WKB reading, independently of any physical identification.

Common origin. The three identifications P1–P3 share a structural common origin: the bifurcation q_+/q_- of the geometric-law parameter at the fixed point $\mu^* = 15$. This bifurcation produces, within PT, the same structure $s^2/2$ that appears both as scalar self-coupling (physical side of the bridge) and as Maslov phase (spectral side). Identity K2 asserts the *uniqueness* of this origin: the three manifestations are projections of the same PT object onto three different axes (arithmetic, physical, spectral).

Status. The K2 identification is proved *structurally*: each of the three sub-identities P1, P2, P3 is exact, and their numerical coincidence at the fixed point makes K2 a *plausible* PT identity in the sense of the program. The cohomological derivation (item K3 of the program [12], in progress) would provide a unified proof of all three, from a single cohomological invariant of the involution $D : q_+ \leftrightarrow q_-$.

4.3 Trace identity A6

Theorem 4.2 (Exact trace identity in $\Re(s) > 1$). *Let $R(s)$ denote the residual factor of a Fredholm representation of the zeta function regularised by PT, and let A_p be the amplitude of the channelled operator at prime p , $A_p = \sin^2 \theta_p(q_+) + \sin^2 \theta_p(q_-)$. In the region $\Re(s) > 1$ one has the exact analytic identity*

$$\boxed{-\frac{d}{ds} \log R(s) = \sum_p \log p \cdot \left[\frac{1}{p^s - 1} + A_p \cdot p^{-s} \right]}. \quad (12)$$

Numerical verification. Identity (12) has been tested numerically (multi-precision `mpmath`, 25 significant digits) on 17 values of s distributed in $\Re(s) > 1$; the gap between the two sides lies at the machine-precision level, i.e. of order 10^{-25} . For all practical purposes, the identity is *exact* [12, Note 62].

Reading. The first term $\sum_p \log p / (p^s - 1)$ is the standard logarithmic derivative of Riemann’s zeta function, $-\zeta'(s)/\zeta(s)$. The second term, $\sum_p \log p \cdot A_p \cdot p^{-s}$, is the *proper PT contribution*: it encodes the effect of the q_+/q_- bifurcation on the regularised trace. Identity (12) therefore asserts that the logarithmic derivative of the residual factor $R(s)$ decomposes cleanly into a “classical zeta background” and a “PT correction” of amplitude A_p .

4.4 Consequence: direct link $s = 1/2 \leftrightarrow$ spectral phenomenology

The two identities K2 and A6 together provide a concrete *arithmetic–spectral bridge*:

- identity K2 (4.1) relates an *observable quantity* (λ_H , measurable at the LHC) to a *spectral quantity* (Δ_N^{Maslov} , predictable from the Riemann zeros) by passing through a *pure PT quantity* ($s^2/2 =$ square of the unique formal input of the theory);

- the trace identity A6 (4.2) provides the *analytic plumbing* that transports the PT amplitudes A_p into the logarithmic derivative of a residual function close to ζ , in the regime $\Re(s) > 1$. These two results are, strictly speaking, *proved identities* (and not unconditional theorems in the sense of 2 and 3): K2 rests on the structural coincidence of three sub-identities, A6 is analytic in $\Re(s) > 1$ but its extension to the critical line remains open. Precisely this extension is the subject of the spectral-arithmetic program presented in 5.

5 Spectral-arithmetic program: the RH-PT conjecture

The identities K2 (4.1) and A6 (4.2) are the cornerstones of a more ambitious research program, in progress [12], whose horizon is a PT-native approach to the Riemann Hypothesis. This section sketches the program, rigorously distinguishing what is proved, what is conjectural, and what remains open. *None* of the results of this section is claimed as an unconditional theorem.

5.1 Residual survival operator

For each prime p and each complex s with $\Re(s) > 1$, define the residual amplitude

$$\kappa_p(s) = 1 - (1 - p^{-s}) \exp(A_p p^{-s}),$$

where $A_p = \sin^2 \theta_p(q_+) + \sin^2 \theta_p(q_-)$. The residual regularised PT zeta function is written as a modified Euler product,

$$R(s) = \prod_p \frac{1}{1 - \kappa_p(s)}, \quad \Re(s) > 1,$$

which converges absolutely in the half-plane $\Re(s) > 1$. The logarithmic derivative of R produces precisely the A6 trace identity ((12)). One then defines the *residual survival operator* of PT as the diagonal operator on $\ell^2(\mathcal{P})$ with spectral symbol $\lambda_p(s) = p^{-s}$; this operator plays, in the PT framework, a role analogous to the semi-classical Hamiltonian $H_{BK} = xp$ of Berry–Keating [14].

5.2 Analytic status and critical boundary

Theorem. The Schur–Fredholm identity associated with the residual survival operator is proved as an *exact operatorial identity* on the Hilbert space $\ell^2(\mathcal{P})$ in the regime $\Re(s) > 1$ [12].

Identity A6. The logarithmic derivative of $R(s)$ admits the exact decomposition into two components ($-\zeta'/\zeta + \text{PT correction } A_p p^{-s}$) in $\Re(s) > 1$, verified numerically to 10^{-25} (theorem 4.2).

Identity K2. The Maslov phase of the semi-classical quantisation of the Riemann zeros coincides with $s^2/2 = 1/8$, triple spectral-physical-arithmetic identification (theorem 4.1).

Conjectural. The analytic continuation of identity (12) to the critical line $\Re(s) = 1/2$ is open. The first edge conditions (Hadamard regularisation, Abel weights, Gaussian smoothing) have been tested numerically [12] and reproduce the position of the Riemann zeros to all available precisions; but no rigorous analytic proof on the critical line is known to date.

5.3 The RH-PT conjecture

Conjecture 5.1 (RH-PT, spectral-lock conjecture). The distributional distribution of poles of the regularised trace

$$\mathcal{T}(s) := \text{Tr}_{\text{reg}}[(s - i H_{BK})^{-1} \cdot D_R(s)]$$

on the critical line $\Re(s) = 1/2$ coincides *exactly* with the set of non-trivial zeros of Riemann’s ζ function.

The RH-PT conjecture has been tested numerically on 17 zeros, over 12 decades, in multi-precision [12, Notes 66–69]. At all available precisions the coincidence is verified, but the epistemic status remains *conjectural*: a rigorous proof on the critical line remains to be provided.

5.4 What the program would deliver

If the RH-PT conjecture were proved, several classical and new results would follow:

1. the Riemann Hypothesis itself, as a consequence of the fact that the poles of $\mathcal{T}$ on the critical line would be in bijective correspondence with the zeros of ζ ;
2. a structural explanation of the position $\Re(s) = 1/2$: the critical line would appear as the *spectral locus* where the PT operator H_{BK} admits the canonical regularisation;
3. a precise operatorial framework for the Hilbert–Pólya program, in which the role of the self-adjoint operator is played by the PT residual survival operator.

Caveat. The RH-PT program belongs to date to *open research*, and its inclusion in this article does not in any way prejudice its resolution. The theorems of sections 2 and 3, as well as the identities K2 and A6 of section 4, are *independent* of the RH-PT conjecture and remain valid as such. The present article proposes RH-PT as a *program horizon*, not as an established result.

6 Conclusion and perspectives

Synthesis of the three readings. The arithmetic cascade of the PT active primes $\{3, 5, 7\}$, fixed by the fixed point $\mu^* = 15$, admits three canonical realisations of complementary natures.

Algebro-geometric reading (2). The cascade is the unique algebraic curve in the Kontsevich–Norbury class satisfying conditions (A) polynomial holonomy, (B) activation threshold, and (C) self-consistency. Its genus is $g(\mathcal{C}_{\text{per}}) = \mu^* - 1 = 14$.

Differential reading (3). The Fisher–Bianchi metric g_{PT} induced by the cascade satisfies the asymptotic Ricci soliton equation $\text{Ric} + \text{Hess}(f) = \lambda(\mu) g_{\text{PT}}$ with $\lambda(\mu) = -1/\mu^4 + O(1/\mu^5)$, coefficient -1 exact.

Spectral reading (4). The axiom $s = 1/2$ manifests in the Riemann–von Mangoldt residue via the triple identity $1/8 = s^2/2 = \lambda_H = \Delta_N^{\text{Maslov}}$ (K2), and the logarithmic derivative of the residual factor admits the exact decomposition $-d/ds \log R(s) = \sum_p \log p [1/(p^s - 1) + A_p p^{-s}]$ verified to 10^{-25} in $\Re(s) > 1$ (A6).

Differentiated epistemic statuses. The results of this article fall into three categories that should be carefully distinguished.

1. *Theorems proved unconditionally:* the classification theorem and the genus formula (2.2, 2.4); the Ricci quasi-soliton theorem (3.1).
2. *Proved identities:* K2 (4.1) is proved structurally as the exact coincidence of three independent sub-identities; A6 (4.2) is an exact analytic identity in $\Re(s) > 1$, verified numerically to all available precisions.
3. *Conjectural program:* the RH-PT conjecture (5.1) is an open research program, under investigation [12]. Its eventual resolution would entail the Riemann Hypothesis.

Articulation with the companion article. This article presupposes, for its full intelligibility, the reading of [1], where the seven structural theorems **T0–T6** and the three proved bridge axioms **BA3–BA5** are established. The geometric and spectral readings developed here are non-trivial extensions: they do not modify the arithmetic core of PT, but reveal its cohomological, differential, and spectral facets. A reverse reading (from this article to Article 1) is also possible: the topological genus 14 and the soliton equation provide *a posteriori* constraints on the cascade that reinforce the uniqueness of the active-prime choice $\{3, 5, 7\}$.

Perspectives. Three immediate research directions emerge from this article.

- *Exhaustive computation of the invariants $\omega_{g,n}^{\mathcal{C}_{\text{per}}}$.* The Eynard–Orantin topological recursion allows, in principle, the computation of all invariants $\omega_{g,n}^{\mathcal{C}_{\text{per}}}$ of the persistence curve. The first computations ([2, §8]) confirm the structural substitution $2\pi^2/3 \rightarrow (1/2) \sum \gamma_p$ relative to Mirzakhani’s Airy curve. Extending this computation to $\omega_{2,1}^{\mathcal{C}_{\text{per}}}$ and beyond would test new cascade identities.

- *Exact identity of the scale bridge* $\lambda \leftrightarrow B_0$. The relation $\lambda(\mu) \sim B_0(\mu)^{-8/3}$ is verified numerically to 0,4 %. The *exact* identity (up to a coefficient) between the Ricci residue λ and the leading coefficient B_0 of topological recursion would constitute a strong algebro-differential bridge.
- *Cohomological promotion of K2*. The derivation of identity K2 from a cohomological invariant of the involution $D : q_+ \leftrightarrow q_-$ (item K3 of the program [12]) would provide a unified proof of the three sub-identities, and a rigorous framework for their extension to dimensions other than $\mathbb{R}^4$.

Links to other PT applications. The readings presented here are in active dialogue with several domains of PT phenomenology, in particular cosmology (the Fisher–Bianchi metric is the PT analogue of Bianchi I, cf. [16, 15]), high-energy physics (the coupling λ_H identified in K2 is the LHC-measurable Higgs self-coupling, cf. [15]), and analytic number theory (the RH-PT conjecture relates the PT survival operator to the Riemann zeros). None of these applications is developed here; they constitute the *downstream* horizon of the present article.

Editorial status. The pair of articles [1]–present, together with the foundational series M1–M5, constitutes, to the author’s knowledge, the most consolidated input material for an external peer review of the Theory of Persistence. The theorems T0–T6, the bridge axioms BA0–BA5, the derivation of α_{EM} , the persistence curve, the Ricci quasi-soliton, and the spectral identities K2 and A6 are presented here with their precise epistemic status. The open questions (RH-PT program, exact identity $\lambda \leftrightarrow B_0$, cohomological promotion of K2) are explicitly flagged and constitute the immediate research horizon of the program.

A Geometric-spectral glossary

Table 1 summarises the geometric and spectral notations specific to this article and their PT reading.

Notation	Definition	PT reading
$\mathcal{C}_{\text{per}}$	Persistence algebraic curve in $\mathcal{KN}$	Unique solution of $\text{ACA}_{\mathcal{P}}$ with $ S \geq 2$
$g(\mathcal{C}_{\text{per}})$	Genus of $\mathcal{C}_{\text{per}}$	$\mu^* - \lfloor S /2 \rfloor = 14$
$\omega_{g,n}^{\mathcal{C}_{\text{per}}}$	Eynard–Orantin invariants	Cascade of PT correlators
g_{PT}	Fisher–Bianchi metric	Canonical Riemannian metric on the μ space
$\text{Ric}(g)$	Ricci tensor of g	Differential curvature of g_{PT}
$\text{Hess}(f)$	Hessian of f	Hessian of the potential $f = -\ln \alpha_{\text{sieve}}$
$\lambda(\mu)$	Ricci residue	$-1/\mu^4 + O(\mu^{-5})$, dynamical signature
$H_p(\mu)$	Local Hubble rate	Rate of change of θ_p with μ
$N(T)$	Riemann-zero count	Riemann–von Mangoldt law
Δ_N^{Maslov}	Maslov phase of the corners	RvM spectral residue 1/8
λ_H	Higgs self-coupling	Scalar coupling at cascade depth $d = 3$
$R(s)$	Regularised zeta residual factor	$\prod_p 1/(1 - \kappa_p(s))$, converges in $\Re(s) > 1$
$\kappa_p(s)$	Channel amplitude at prime p	$1 - (1 - p^{-s}) \exp(A_p p^{-s})$
A_p	PT amplitude at p	$\sin^2 \theta_p(q_+) + \sin^2 \theta_p(q_-)$
H_{BK}	Regularised Berry–Keating Hamiltonian	PT operator xp regularised

Table 1: Geometric-spectral glossary of Article 2.

B Summary table of results

Table 2 synthesises the principal results of this article, classified by epistemic status with their references.

Label	Short statement	Section	Status
Theorems proved unconditionally			
Th. Classification (2.2)	$S = \{3, 5, 7\}$ unique ACA $_{\mathcal{P}}$ solution with $ S \geq 2$	2	[THM]
Th. Algebraicity (2.3)	$\mathcal{C}_{\text{per}}$ algebraic, $Y^2 \in \mathbb{Q}(x)$	2	[THM]
Th. Genus (2.4)	$g(\mathcal{C}_{\text{per}}) = \mu^* - \lfloor S /2 \rfloor = 14$	2	[THM]
Th. Quasi-soliton (3.1)	$\lambda(\mu) = -1/\mu^4 + O(\mu^{-5})$, coef. -1 exact	3	[THM]
Prop. Anisotropy (3.2)	Anisotropy $\Leftrightarrow$ uniqueness of λ	3	[THM]
Proved identities			
K2 (4.1)	$1/8 = s^2/2 = \lambda_H = \Delta_N^{\text{Maslov}}$	4	[ID] structural
A6 (4.2)	$-d/ds \log R = \sum_p \log p [1/(p^s - 1) + A_p p^{-s}]$	4	[ID], $\Re(s) > 1$
Conjectural program			
RH-PT (5.1)	Poles of $\mathcal{T}(s)$ on $\Re(s) = 1/2 \leftrightarrow$ zeros of ζ	5	[CONJ]
$\lambda \leftrightarrow B_0$	$\lambda(\mu) \sim B_0(\mu)^{-8/3}$ exact	3.5	[CONJ]
K3 cohomological	Unified cohomological derivation of K2	4.2	[OPEN]

Table 2: Summary table of the article’s results. [THM] = unconditional theorem, [ID] = proved identity, [CONJ] = conjecture, [OPEN] = open problem.

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